Appendix: Diagnosing Memory-Write Decisions in Conversational Assistants Through Specification-Based Replay

Contents and scope

This supplement accompanies the manuscript *Diagnosing Memory-Write Decisions in Conversational Assistants Through Specification-Based Replay*. Condition names follow the main article. AGREE, ALLOW, LITERAL, and WITNESS are the admission rules. REVIEW is the complete pipeline compared in the main article: a draft plus one general-review call. The original protocol also ran two further complete pipelines that the main article mentions only briefly: DIRECT, which executes the direct draft, and REFINE, a single draft–feedback–refine cycle adapted from Self-Refine; both share REVIEW's initial draft where the design permits. The controlled path combines extraction, proposal, and WITNESS admission; WITNESS-R denotes the same path with its reply rendered from the execution log, which is the arm of the internally prespecified primary test. Scores use the main article's specification-based definitions, and "Viol." abbreviates reference-violating states. Case identifiers such as CT45 refer to the archived test cases.

Here, "prespecified" means fixed in the original internal protocol; the study was not publicly preregistered. Diagnostic follow-ups were informed by earlier results and fixed before their own outputs. Model configurations and repetitions increase observations, not independent cases. The exact reference and the mechanical variant-tolerant reference have not been independently validated by external annotators. The reported evidence includes no independent human annotation or external-writer replication.

A Counts of the original protocol for Gemini 2.5 Flash and Llama 3.1 8B.
B Llama 3.1 8B and the eight-configuration rule and pipeline profiles.
C Model roster, generation settings, and recorded resources.
D Contrast set.
E Boundary probe.
F Public task-dialogue probe.
G Complete three-repetition model panel, selection record, repetition stability, and controlled-path exposure.
H Translated examples of corrections and held writes.
I Admission bounds, operation split, and variant audit on saved REVIEW proposals.
J Follow-up runs: revised extraction instruction, learned verifier, real speech-recognition hypotheses (DSTC2), and the status of a Korean memory-operation source.
K Variant-tolerant re-scoring: definition, accepted variants, rule comparisons and configurations under both references, and the decomposition of the complete-pipeline difference.

A Counts of the Original Protocol

A.1 Units and Model Settings

The original protocol fixed 48 held-out cases, three repetitions, and seven conditions per model configuration: DIRECT, REVIEW, REFINE, AGREE, WITNESS, and the reply-rendered variants of AGREE and WITNESS. Because the conditions share six generation stages, each completed run has 1,008 scored condition rows but 864 physical model calls; condition rows, model calls, and cases are therefore different units. Positive-change completion uses 28 cases (84 observations per condition), strict correction completion 11 cases (33 observations), and negative preservation the remaining

20 cases (60 observations). The 12 development cases are excluded from every score, and each completed run also passed 12 technical probe calls that are reported separately from the held-out calls.

Gemini 2.5 Flash used the Vertex model `gemini-2.5-flash` with temperature 0.2 and thinking disabled. Llama 3.1 8B used the local four-bit snapshot mlx-community/Llama-3.1-8B-Instruct-4bit (revision `90215b22ec18e72f623dde2ea7af4097025160e2`) with base seed 20260927. No model parameters were trained.

A.2 Reference and Endpoints

A fact is an exact subject–relation–value–time tuple. The authored reference lists the allowed actions, required and forbidden tuples, the record identifiers whose facts must remain unchanged, the identifiers to delete, and the obsolete versions to retire, so a correction may keep its record identifier while replacing an obsolete tuple. This retirement distinction was added to the evaluator during development and fixed, together with the test reference, before held-out collection. The two co-first authors independently checked each of the 48 held-out references against its case; no agreement statistic was computed. Whole-state conformity requires a valid final output, every required fact, no forbidden or obsolete tuple, preserved protected facts, the specified removals, and no unlisted fact. Positive-change completion additionally requires an allowed action on a positive case, and strict correction completion applies the same criterion to the correction-eligible cases. A reference-violating state is a forbidden or extra write or damage to a protected fact; it can coincide with other failures. Invalid outputs remain in the planned denominators.

A.3 Condition Counts

Tables A1–A3 give the complete counts for the two original-family configurations that completed first. The reply-rendered variants of Agree and Witness have the same counts as Agree and Witness on every endpoint, because rendering does not change the executed state; the tables therefore list each admission rule once. Qwen3 14B, the third completed original family, is reported in the main article and in Appendices G and I.

Table A1. Endpoint counts of the original protocol. Positive-change completion uses 84 observations, strict correction 33, and whole-state conformity, reference-violating states, and invalid outputs 144, corresponding to 28, 11, and 48 cases with three repetitions.

Configuration	Condition	Positive /84	Correct. /33	Conform. /144	Viol. /144	Invalid /144
Gemini 2.5 Flash	Direct	81	32	137	6	1
	Review	82	33	138	5	1
	Refine	73	32	131	10	2
	Agree	69	22	128	3	1
	Witness	80	33	139	3	1
Llama 3.1 8B	Direct	24	0	82	14	2
	Review	29	4	84	17	1
	Refine	25	2	84	19	0
	Agree	6	0	30	24	68
	Witness	6	0	30	24	68

Table A2. Secondary counts of the original protocol. Negative preservation uses the 20 cases that require no change (60 observations). The clarification and extra-write columns count reference flags over all 144 observations per condition, not rates conditional on need; extra writes and reference-violating states overlap and should not be added.

Configuration	Condition	Negative	Missed clarif.	Unneeded clarif.	Extra write
Gemini 2.5 Flash	Direct	56	1	1	6
	Review	56	1	1	5
	Refine	58	0	1	10
	Agree	59	0	1	3
	Witness	59	0	1	3
Llama 3.1 8B	Direct	55	3	44	14
	Review	55	9	1	17
	Refine	54	3	32	19
	Agree	18	2	56	19
	Witness	18	2	56	19

Table A3. Whole-state conformity by problem family, out of 12 observations (four cases, three repetitions) per cell. Totals equal the conformity counts in Table A1.

Family	Direct	Review	Refine	Agree	Witness
Gemini 2.5 Flash					
Explicit correction	12	12	12	10	12
Negated correction	12	11	12	12	12
Uncertain report	12	12	12	12	12
Wrong subject	12	12	6	12	12
Temporal distinction	12	12	12	12	12
Transcript conflict	11	11	11	11	11
Equivalent variants	12	12	11	11	11
New append	10	10	9	9	9
Permission unknown	12	12	11	12	12
Stop or decline	9	10	12	12	12
Partial scope	11	12	12	11	12
Self-repair	12	12	11	4	12
Total /144	137	138	131	128	139
Llama 3.1 8B					
Explicit correction	0	0	1	0	0
Negated correction	12	12	12	7	7
Uncertain report	12	11	12	6	6
Wrong subject	0	0	0	1	1
Temporal distinction	5	5	7	7	7
Transcript conflict	8	8	6	3	3
Equivalent variants	9	9	6	4	4
New append	8	8	8	0	0
Permission unknown	12	12	12	2	2
Stop or decline	11	12	12	0	0
Partial scope	4	6	6	0	0
Self-repair	1	1	2	0	0
Total /144	82	84	84	30	30

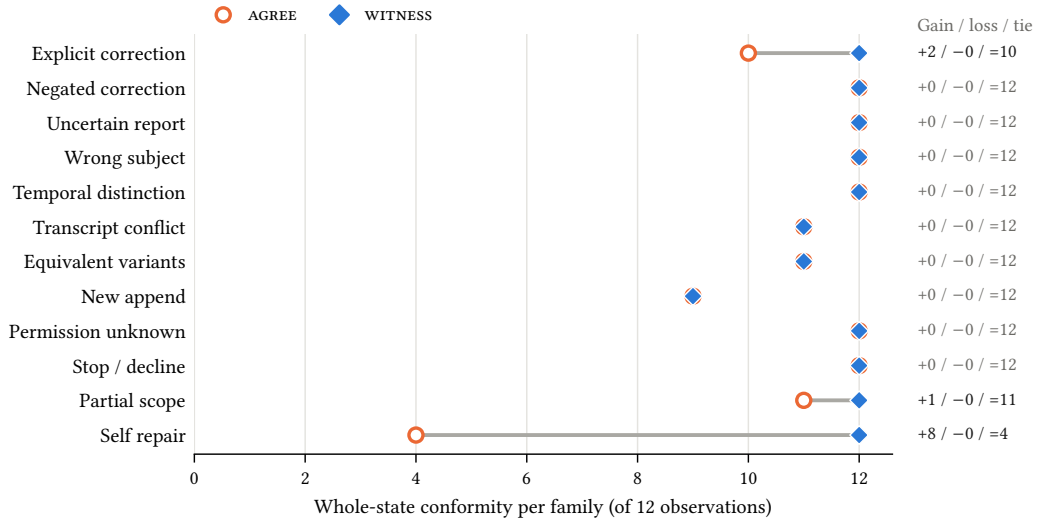


Fig. A1. Recovery by problem family on Gemini 2.5 Flash. The 11 observation-level gains of Witness over Agree come from five correction cases, and eight of them occur in self-repair.

A.4 Statistics and Held Corrections

The internally prespecified primary comparison is Witness-r minus Review on positive-change completion. Rates average the three repetitions within each eligible case, and the 95% percentile interval uses 5,000 fixed-seed case-cluster bootstrap draws. The exact McNemar test first reduces each condition to a majority of three successes per case. The five planned model families keep their Holm accounting: three slots have observed tests (Gemini 2.5 Flash, Llama 3.1 8B, and Qwen3 14B), and the two uncollected slots (Appendix C) enter with an adjustment value of one and no observed p. The Agree–Witness contrasts, family profiles, and failure counts are descriptive.

On Gemini 2.5 Flash, Agree held 11 required corrections that Witness admitted: all three repetitions of CT45 and CT46, two of CT01 and CT48, and one of CT42. Table A3 and Figure A1 show eight of them in the self-repair family. All reported runs are complete, and no failed output was regenerated. Five Gemini calls ended in rate-limit errors and lack token usage; the other 859 Gemini calls and all 864 Llama calls report usage.

B Llama 3.1 8B and Eight-Configuration Profiles

Llama 3.1 8B belonged to the original comparison and was not selected after its result. Agree and Witness each complete 0 of 33 strict corrections and 6 of 84 positive changes, against 29 for Review, 24 for Direct, and 25 for Refine. The Witness-r–Review case-mean difference is −27.4 percentage points (95% case-cluster interval −44.0 to −10.7), with exact majority-case McNemar $p = .0117$ and original five-slot Holm-adjusted $p = .0586$. All 864 provider calls returned, but the controlled path produced 68 invalid final outputs out of 144, against one for Review, two for Direct, and none for Refine, and its valid outputs also contain action and state failures. Model family, size, four-bit quantization, runtime, and output-schema demands vary together in this configuration, so the result does not isolate parameter count.

Only two Llama outputs reach H. In CT47, repetition 1, Witness keeps a hold on an eligible correction that Allow releases. Overall, Allow produces one more reference-violating state than Agree, Literal, and Witness (25 versus 24), and strict correction completion remains zero under

every rule (Table B1). Table B2 sets the complete pipelines of the same eight configurations side by side; these pipelines generate different proposals.

Table B1. Admission rules on identical proposals for the eight configurations of the original protocol and the Gemini series, in the order AGREE / ALLOW / LITERAL / WITNESS. The Gemini rows repeat Table 6 of the main article. *H* counts outputs that reach the held set.

Configuration	Correction /33	Viol. /144	*H* outputs
Gemini 2.5 Flash	22 / 33 / 33 / 33	3 / 3 / 3 / 3	11
Gemini 2.5 Pro	31 / 31 / 31 / 31	3 / 9 / 4 / 3	6
Gemini 2.5 Flash-Lite	31 / 31 / 31 / 31	16 / 19 / 16 / 16	3
Gemini 3 Flash Preview	33 / 33 / 33 / 33	12 / 12 / 12 / 12	0
Gemini 3.1 Pro Preview	33 / 33 / 33 / 33	2 / 2 / 2 / 2	0
Gemini 3.5 Flash-Lite	32 / 32 / 32 / 32	7 / 12 / 7 / 7	5
Gemini 3.8 Flash	33 / 33 / 33 / 33	1 / 1 / 1 / 1	0
Llama 3.1 8B	0 / 0 / 0 / 0	24 / 25 / 24 / 24	2

Table B2. Complete pipelines for the same eight configurations, in the order DIRECT / REVIEW / controlled path. Counts combine three repetitions within each configuration and are not pooled across configurations.

Configuration	Correction /33	Viol. /144
Gemini 2.5 Flash	32 / 33 / 33	6 / 5 / 3
Gemini 2.5 Pro	33 / 32 / 31	8 / 8 / 3
Gemini 2.5 Flash-Lite	33 / 33 / 31	14 / 14 / 16
Gemini 3 Flash Preview	33 / 33 / 33	6 / 6 / 12
Gemini 3.1 Pro Preview	33 / 33 / 33	3 / 3 / 2
Gemini 3.5 Flash-Lite	33 / 33 / 32	1 / 0 / 7
Gemini 3.8 Flash	33 / 33 / 33	1 / 0 / 1
Llama 3.1 8B	0 / 4 / 0	14 / 17 / 24

C Model Roster, Generation Settings, and Resources

C.1 Roster

The original protocol planned a Holm-adjusted primary comparison for five model families. Gemini 2.5 Flash, Llama 3.1 8B, and Qwen3 14B (local 4-bit) completed. EXAONE 3.5 7.8B stopped at the technical probe for incomplete coverage, and a planned 12B configuration was replaced before collection, so neither contributes a result (Table C1). An earlier version of this analysis excluded the Qwen3 14B results after they had become available; they are reported in full here and in the main article, and the five-slot Holm accounting is the one fixed before collection. The later Gemini configurations reuse the 48 cases, are descriptive, and do not enter the Holm family. "Complete" means that every scheduled main-source call and the first analysis finished; a technical stop is not a zero score. Appendix G records how the 21-configuration panel was selected.

The original call budget was $48 \times 3 \times 6 = 864$ main-source calls plus 12 technical probes per configuration; the later Gemini runs separately allowed 864 main-source, 96 contrast-set, and 12 probe calls, with no retry. These budgets are not effective sample sizes: every complete configuration shares the same 48 authored cases.

Table C1. Model roster of the original protocol and the Gemini series.

Model	Design block	Main-source status
Gemini 2.5 Flash	Original family	Complete
Llama 3.1 8B	Original family	Complete
Qwen3 14B (4-bit)	Original family	Complete
EXAONE 3.5 7.8B	Original family	Stopped at technical probe
Planned 12B configuration	Original family	Replaced before collection
Gemini 3.8 Flash	Later addition	Complete
Gemini 2.5 Pro, 2.5 Flash-Lite, 3 Flash Preview, 3.1 Pro Preview, 3.5 Flash-Lite	Gemini series	Complete
Gemini 3.5 Flash, 3.6 Flash, 3.7 Flash	Gemini series	Stopped by provider errors
Gemini 3.1 Flash-Lite	Gemini series	Stopped at technical probe

C.2 Generation Settings

Table C2 lists the requested settings for the seven Gemini configurations and Llama 3.1 8B. Requested temperature and thinking settings are not measurements of effective sampling, and returned cloud version strings identify service names rather than immutable weights.

Table C2. Requested generation settings. Caps are maximum output tokens for ordinary stages / feedback. Thinking numbers are requested token budgets, and LOW is the requested thinking level. Version rows count calls whose returned version string matches the requested identifier.

Requested model identifier	Caps	Temperature	Thinking	Version rows
`gemini-2.5-flash`	2048/512	0.2	0	859/864
`gemini-2.5-pro`	8192/2048	0.2	128	853/864
`gemini-2.5-flash-lite`	8192/2048	0.2	0	851/864
`gemini-3-flash-preview`	8192/2048	0.2	LOW	861/864
`gemini-3.1-pro-preview`	8192/2048	0.2	LOW	855/864
`gemini-3.5-flash-lite`	8192/2048	provider default	LOW	856/864
`gemini-3.8-flash`	8192/2048	0.2	LOW	864/864
`Llama-3.1-8B-Instruct-4bit`	2048/512	0.2	–	864/864

Cloud calls used a REST adapter written in Python, and Llama 3.1 8B used MLX adapter code with a fixed runtime path; its adapter records the model identifier as the version rather than a version returned by a server.

C.3 Recorded Resources

Each configuration made 864 physical calls, including failures; across the eight configurations, 6,863 of 6,912 calls returned. Token usage is missing for the 49 failed calls, and summed latency is not elapsed study time (Table C3). Table C4 isolates the two-call Review pipeline and the controlled path. Witness uses the same two saved calls per output as Agree, so evaluating it, like Allow and Literal, adds no model call.

Table C3. Recorded physical calls for the seven Gemini configurations and Llama 3.1 8B. Token sums use the calls that report a value; reasoning tokens are reported separately where present. Latency is a sum of per-call seconds, not wall-clock time.

Configuration	OK	Failed	Input	Output	Reasoning	Usage	∑ s
Gemini 2.5 Flash	859	5	1,546,070	189,292	0	859/864	1441.6
Gemini 2.5 Pro	853	11	1,542,438	201,837	76,052	853/864	3493.6
Gemini 2.5 Flash-Lite	851	13	1,575,962	230,385	0	851/864	1289.8
Gemini 3 Flash Preview	861	3	2,301,473	126,385	43,639	861/864	2804.3
Gemini 3.1 Pro Preview	855	9	2,267,984	170,045	27,803	855/864	3302.1
Gemini 3.5 Flash-Lite	856	8	2,273,983	146,597	0	856/864	1316.2
Gemini 3.8 Flash	864	0	2,294,575	115,377	2,271	864/864	3019.5
Llama 3.1 8B	864	0	1,570,636	161,305	–	864/864	9408.9

Table C4. Saved calls behind the Review pipeline and the controlled path. Each covers 144 outputs and 288 physical calls; these sums overlap with Table C3 and should not be added to it.

Configuration	Path	Calls	Input	Output	∑ s
Gemini 2.5 Flash	Review	288	512,062	54,918	431.9
	Controlled	288	533,378	77,022	519.4
Gemini 2.5 Pro	Review	288	508,872	57,657	1088.7
	Controlled	288	533,190	89,653	1302.2
Gemini 2.5 Flash-Lite	Review	288	509,749	58,254	378.7
	Controlled	288	533,820	82,205	427.1
Gemini 3 Flash Preview	Review	288	821,829	40,476	877.7
	Controlled	288	804,384	54,497	1093.1
Gemini 3.1 Pro Preview	Review	288	809,106	54,048	1082.7
	Controlled	288	795,470	85,360	1197.4
Gemini 3.5 Flash-Lite	Review	288	817,567	43,449	417.2
	Controlled	288	787,723	70,089	494.6
Gemini 3.8 Flash	Review	288	823,976	36,927	955.8
	Controlled	288	801,856	57,104	1289.4
Llama 3.1 8B	Review	288	525,149	49,177	3030.1
	Controlled	288	537,723	69,668	3635.4

D Contrast Set

The authors specified the 24-case contrast set after the initial recovery finding, and a separate AI agent drafted it from the schema without the rule code or earlier outputs. It pairs 12 supported and 12 unsupported edits with matched records and task instructions, and its references were drafted in the same way and fixed before execution. The set was run once on Gemini 2.5 Flash (Vertex, temperature 0.2, thinking budget 0, at most 2,048 output tokens, no retries). Case labels and expected states were withheld from model messages; branch order was counterbalanced by case identifier, and case order was shuffled with seed 2026092720. Each case used two extraction-and-proposal calls shared by the admission rules and two separate Review calls: 96 calls, 95 of which returned; one draft-stage rate-limit error was followed by the scheduled final call.

Because the agent worked without the rule code or earlier outputs, the set tests the diagnosis on cases written independently of our rule design; external human validation of the set is planned (main article, Section 9).

The admission rules received identical saved proposals, executed on isolated SQLite states initialized from the same case. All 24 extraction-and-proposal paths were valid, 18 cases contained a proposed CORRECT, and AGREE held six for lack of shared support although every reading was validly covered. WITNESS restored five, each with affirmed, target-linked witnesses, and kept one hold because a witness was not affirmed; LITERAL and ALLOW restored all six. The scorer compares persistent record identifiers, subjects, relations, values, and time fields.

On supported cases, paired gains and losses are 5/0 for WITNESS versus AGREE, 0/0 versus ALLOW, and 0/2 versus REVIEW; on unsupported cases they are 0/0, 1/0, and 2/1. The corresponding exact McNemar p values are .0625, 1, .5, 1, 1, and 1; with one draw per case and no multiplicity adjustment, they describe these cases only (Table D1).

Table D1. Whole-state outcomes on the contrast set with 95% Wilson intervals; each stratum has 12 cases.

Condition	Supported edit completed	Unsupported edit preserved
AGREE	5/12, 41.7% [19.3, 68.0]	9/12, 75.0% [46.8, 91.1]
ALLOW	10/12, 83.3% [55.2, 95.3]	8/12, 66.7% [39.1, 86.2]
LITERAL	10/12, 83.3% [55.2, 95.3]	8/12, 66.7% [39.1, 86.2]
WITNESS	10/12, 83.3% [55.2, 95.3]	9/12, 75.0% [46.8, 91.1]
REVIEW	12/12, 100.0% [75.8, 100.0]	8/12, 66.7% [39.1, 86.2]

Table D2. Translated trace of case N015, the only contrast-set case in which WITNESS and ALLOW differ. The two readings are alternatives for the same utterance, not two speakers.

Stage	Saved content or executed result
Stored record	f1: participant, residence, Ulsan, present.
Task instruction	Edit only the current record. Add no other-person or other-time facts. Keep the record when the meaning is unresolved.
Reading 1	I live in Changwon now.
Reading 2	I do not live in Changwon now.
Shared proposal	CORRECT f1: residence ← Changwon.
Shared reply	You live in Changwon. Understood.
Witnesses	Reading 1 affirmed; reading 2 negated.
AGREE	Held for lack of shared support; Ulsan retained.
WITNESS	Hold kept because a witness is not affirmed; Ulsan retained.
LITERAL and ALLOW	Hold released; Changwon written.
REVIEW	Changwon written from its separate proposal.

The kept hold prevents the disputed overwrite, but the unchanged reply still asserts the disputed value, which separates protection of the stored state from the quality of the reply (Table D2). The other unsupported-case failures of WITNESS and ALLOW keep the target but append facts outside the permitted edit scope. On supported cases, both rules replace all 12 target values, and two additional past-state records fail the whole-state endpoint.

E Boundary Probe

The 12-case boundary probe was run once with the same Gemini 2.5 Flash configuration and rule implementations, using 24 model calls and producing 36 condition rows rather than 36 cases. Its source and reference were drafted to the authors' specification with an AI tool and fixed before collection. Six pairs share an initial record and vary whether the current utterance licenses updating

it. Because no output reached H, all four admission rules produce the same state in every case (Table E1).

Table E1. Boundary-probe outcomes. A match requires a valid output and the complete expected fact map, so a correct target with an extra fact fails.

Case	Expected edit	Boundary	Whole-state outcome (all rules)
W001	Keep	Hypothetical	Match
W002	Update	Speaker identity	Error: extra past address
W003	Keep	Retracted certainty	Match
W004	Update	Negation	Error: new hobby appended; retired target retained
W005	Keep	Speaker identity	Match
W006	Update	Quotation	Match
W007	Keep	Negation	Match
W008	Update	Retracted certainty	Match
W009	Update	Hypothetical	Match
W010	Keep	Quotation	Match
W011	Update	Temporal scope	Match
W012	Keep	Temporal scope	Error: extra past workplace

For each rule, 4 of 6 supported edits (Wilson 95% interval 30.0 to 90.3%) and 5 of 6 unsupported edits (43.6 to 97.0%) match the reference, and all 24 provider responses and 12 final outputs were valid. W004 shows a limit outside the held set. Both readings state that swimming is the current hobby and hiking has stopped; the proposal appends swimming and issues a CORRECT for hiking that keeps the old target. The hiking witness is negated, but AGREE admits the old tuple because it is already stored, and since the operation is not a correction held for lack of support, WITNESS does not inspect it. In W007, one reading omits a negation about swimming while both continue to support the stored hiking record, and every rule keeps that record.

F Public Task-Dialogue Probe

The source contains 16 selected written user turns from MultiWOZ 2.4, paired with prior task-state subsets. The Gemini 2.5 Flash collection has 96 requests and 144 condition records across nine conditions; a planned run with a second model was dropped before this source was first scored. The reference and eligibility were fixed before collection: 10 same-task self-repairs are eligible, and six plan-change or annotation-ambiguous cases have no automatic correctness label. The reference adapts the dataset's state annotation and is not a new human assessment.

Focal agreement requires exactly one fact for each evaluated subject and relation, equal to the adapted annotation after case folding and whitespace normalization. Unlike positive-change completion, it does not require an allowed action, and generation validity is tracked separately; in the one provider-error case, the unchanged initial value differs from the reference and fails focal agreement.

Table F1. Focal-state agreement on the 10 eligible self-repairs. The reply-rendered variants of AGREE and WITNESS have the same states as AGREE and WITNESS.

	DIRECT	REVIEW	REFINE	AGREE	ALLOW	LITERAL	WITNESS
Agreement /10	10	10	5	3	7	4	4

DIRECT and REVIEW produce identical final states on every source case, as do WITNESS and LITERAL. AGREE holds eight corrections and one append; WITNESS and LITERAL restore four of the held corrections, and ALLOW restores all eight. One WITNESS restoration adds an eligible exact agreement, and ALLOW adds three more (Table F1). A source spelling "Thursday" against a proposed "thursday" shows the cost of literal matching (main article, Section 6.3). In that saved example, the extracted fact list is empty, so normalizing existing facts cannot repair the hold; a lexical diagnostic that applies Unicode NFC, case folding, and whitespace normalization turns the value-to-source and value-to-quotation containment checks from false to true, but that repair was not executed or scored. One candidate-final rate-limit error yields six invalid records across the admission-rule conditions, and two other cases per admission-rule condition contain writes rejected by the executor; no failed request was retried. The source is written task dialogue, not noisy acoustic transcription or personal biography.

G Complete Three Repetition Model Panel

The main panel is selected from 28 configurations with 48 complete case triplets. Seven do not meet the common 95% final-output-validity criterion and six additional archived configurations have incomplete triplets. Table G1 lists all 21 selected configurations, and Figure G1 plots their complete-pipeline differences. The eligibility criterion applies to each whole path over 144 observations; observed invalid outputs remain in the endpoint denominators. Table G2 records the remaining configuration coverage without interpreting uncollected episodes as zero-valued semantic outcomes.

The case-majority McNemar result for Gemini 2.5 Flash AGREE versus WITNESS is $p = .125$, with four discordant cases, and is descriptive. The McNemar test concerns the binarized case-level outcome, whereas the effect size and case-cluster interval in Section 6.1 describe the mean difference. For the original whole-path primary comparison, the case-majority result is $p = 1.00$. The expanded panel is exploratory, and its pointwise intervals do not supply a simultaneous model ranking. Missing pooled intervals remain marked rather than being constructed by combining intervals from different collection phases. Per repetition, the direction of the positive-completion difference changes on 11 of the 21 configurations, counting a tie as a direction (Table G4), and strict correction completion under the controlled path matches REVIEW on 17 configurations and is lower on 4.

G.1 Exposure of Controlled-Path Proposals on All 21 Configurations

Table G5 extends the exposure count of Section 6.2 from the seven Gemini configurations to the whole panel. For every configuration, AGREE and WITNESS are the saved conditions on the controlled-path proposals, and LITERAL is the exact replay executed with the frozen executor as the comparator for normalized LITERAL (Section 6.3); all three use the same extraction and proposal in each configuration, case, and repetition. An output is exposed when AGREE holds at least one proposed CORRECT for lack of shared extracted support while every reading is validly covered. The counts come from the saved gate logs; no output was regenerated or rescored. ALLOW was replayed only for the Gemini configurations (Table 6) and Qwen3 14B (Section 6.2), so its column is omitted.

The unexposed configurations are Claude Opus 4.5, 4.6, 4.7, 4.8, and 5.5, Claude Sonnet 4.6, Gemini 3 Flash Preview, 3.1 Pro Preview, and 3.8 Flash, and GPT 5.6 Luna, Sol, and Terra; on each, the three rules produce identical states. The 35 kept holds are 29 in the transcript-conflict cases CT21, CT22, and CT24 and six in the stop-or-decline cases CT39 and CT40. The two LITERAL-only releases are the Gemini 2.5 Pro workplace overwrite of Section 6.4 (CT15, repetition 2) and a GLM 5 destination overwrite (CT11, repetition 1) that replaces the stored value although the participant

Table G1. All 21 complete configurations, REVIEW → controlled path. Every result combines three repetitions. Positive differences use 28 case clusters. A dash denotes an unavailable matching pooled interval; a partial-collection interval is never substituted.

Configuration	Positive /84	Viol. /144	Invalid /144	Positive Δ pp [95% CI]
Claude Opus 4.5	79 → 78	5 → 6	0 → 0	-1.2 [-10.7, +4.8]
Claude Opus 4.6	76 → 78	8 → 6	3 → 0	+2.4 [-3.6, +10.7]
Claude Opus 4.7	78 → 80	6 → 3	0 → 0	+2.4 [-3.6, +9.5]
Claude Opus 4.8	81 → 82	3 → 2	0 → 0	+1.2 [+0.0, +3.6]
Claude Opus 5.5	69 → 72	15 → 12	0 → 0	+3.6 [+0.0, +9.5]
Claude Sonnet 4	77 → 76	6 → 6	4 → 4	-1.2 [-9.5, +7.1]
Claude Sonnet 4.5	79 → 81	9 → 3	0 → 1	+2.4 [+0.0, +6.0]
Claude Sonnet 4.6	81 → 81	3 → 3	2 → 0	+0.0 [+0.0, +0.0]
DeepSeek 3.2	81 → 76	5 → 7	2 → 2	-6.0 [-14.3, +0.0]
Gemini 2.5 Flash-Lite	69 → 56	14 → 16	1 → 4	-15.5 [-34.5, +4.8]
Gemini 2.5 Pro	71 → 73	8 → 3	3 → 3	+2.4 [-8.3, +11.9]
Gemini 3.1 Pro Preview	81 → 81	3 → 2	3 → 2	+0.0 [-3.6, +3.6]
Gemini 3.5 Flash-Lite	84 → 74	0 → 7	1 → 5	-11.9 [-23.8, -2.4]
Gemini 3.8 Flash	84 → 83	0 → 1	0 → 0	-1.2 [-3.6, +0.0]
Gemini 3 Flash Preview	78 → 71	6 → 12	1 → 0	-8.3 [-19.0, +0.0]
Gemini 2.5 Flash	82 → 80	5 → 3	1 → 1	-2.4 [-6.0, +0.0]
GLM 5	81 → 81	3 → 2	0 → 0	+0.0 [-3.6, +3.6]
GPT 5.6 Luna	77 → 80	7 → 4	0 → 0	+3.6 [—]
GPT 5.6 Sol	71 → 75	13 → 8	0 → 0	+4.8 [+0.0, +10.7]
GPT 5.6 Terra	71 → 76	13 → 5	0 → 1	+6.0 [-1.2, +15.5]
Qwen3 14B (4-bit)	73 → 66	14 → 20	0 → 0	-8.3 [-23.8, +6.0]

asked not to change it; the same case produces the witness-only hold on saved Qwen3 14B REVIEW proposals (Section 6.4).

Table G2. Configuration selection record. All thresholds are technical and use the same rule for both paths.

Configuration	Complete cases per repetition	Valid Review / controlled	Reason outside main panel
Claude Haiku 4.5	48 / 48 / 48	115/144; 96/144	Below 95% valid final outputs
Claude Opus 5	23 / 0 / 0	Incomplete	Fewer than 48 complete case triplets
Claude Sonnet 5	48 / 46 / 0	Incomplete	Fewer than 48 complete case triplets
EXAONE 4 32B	48 / 48 / 48	136/144; 123/144	Below 95% valid final outputs
Llama 3.1 8B	48 / 48 / 48	143/144; 76/144	Below 95% valid final outputs
MiniMax M2.1	48 / 3 / 0	Incomplete	Fewer than 48 complete case triplets
MiniMax M2.5	48 / 40 / 0	Incomplete	Fewer than 48 complete case triplets
Qwen2.5 7B Instruct	48 / 48 / 48	87/144; 72/144	Below 95% valid final outputs
Qwen2.5 32B	48 / 48 / 48	141/144; 123/144	Below 95% valid final outputs
Qwen3 32B	48 / 48 / 48	141/144; 136/144	Below 95% valid final outputs
Qwen3 4B Instruct 2507	48 / 48 / 48	30/144; 51/144	Below 95% valid final outputs
Qwen3 8B	25 / 0 / 0	Incomplete	Fewer than 48 complete case triplets
Qwen3 Coder Next	48 / 25 / 0	Incomplete	Fewer than 48 complete case triplets

Table G3. Sensitivity to final-output validity among configurations with all three repetitions complete. These descriptive counts do not select a threshold for the most favorable result.

Minimum validity	Configurations	Positive higher / lower / same	Reference violations lower / higher / same
100%	9	5 / 3 / 1	6 / 3 / 0
95%	21	9 / 9 / 3	12 / 7 / 2
90%	22	9 / 10 / 3	12 / 8 / 2
No validity filter	28	10 / 15 / 3	16 / 10 / 2

Table G4. Within-panel repetition stability on the same 21 configurations. All main outcome tables use the three-repetition pooled estimates.

Repetition	Configurations	Positive higher	Positive lower	Same
1	21	7	8	6
2	21	6	7	8
3	21	7	7	7

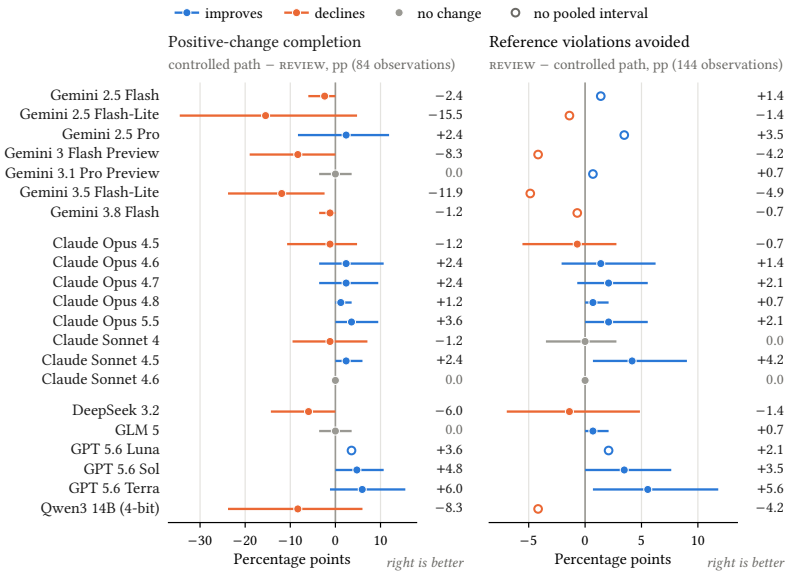


Fig. G1. Effects of the complete pipelines on the same 48 cases over three repetitions, oriented so that improvement is to the right (more positive completions and fewer reference-violating states). Positive completion uses 84 observations from 28 cases, and the reference-violation endpoint uses 144 observations from 48 cases. Blue marks improvement, orange decline, and gray no change; values at the right are point estimates in percentage points. Lines show the matching archived pointwise 95% case-cluster intervals, and open points mark configurations without a complete pooled interval.

Table G5. Controlled-path exposure on the 21 panel configurations. "Released" counts exposed outputs whose held correction both evidence checks restore; every such release completes a required correction. "LITERAL only" counts releases by LITERAL that WITNESS holds; both produce a reference-violating state. "Kept" counts holds that all three rules keep; all 35 lie in cases that require no change, and the held state conforms to the reference in each. The last two columns are in the order AGREE / LITERAL / WITNESS and combine three repetitions; the panel totals use 693 correction and 3,024 all-case observations.

Configuration	*H* **outputs (cases)**	**Rel.**	**LITERAL only**	**Kept**	**Correction /33**	**Viol. /144**
Claude Sonnet 4	3 (CT21)	0	0	3	31/31/31	6/6/6
Claude Sonnet 4.5	4 (CT21, CT22)	0	0	4	33/33/33	3/3/3
DeepSeek 3.2	4 (CT21, CT22, CT46)	1	0	3	32/33/33	7/7/7
Gemini 2.5 Flash	11 (CT01, CT42, CT45, CT46, CT48)	11	0	0	22/33/33	3/3/3
Gemini 2.5 Flash-Lite	3 (CT21)	0	0	3	31/31/31	16/16/16
Gemini 2.5 Pro	6 (CT15, CT21, CT22)	0	1	5	31/31/31	3/4/3
Gemini 3.5 Flash-Lite	5 (CT21, CT22)	0	0	5	32/32/32	7/7/7
GLM 5	3 (CT11, CT21, CT22)	0	1	2	33/33/33	2/3/2
Qwen3 14B (4-bit)	10 (CT21, CT24, CT39, CT40)	0	0	10	33/33/33	20/20/20
12 other configurations	0	–	–	–	equal	equal
All 21	49 (12 cases)	12	2	35	673/685/685	131/133/131

H Detailed Translated Memory Examples

A Opening-year correction
main source

Stored record: shop opening year = 1991

Readings 1 and 2: both correct the year to 1993

Shared proposal: CORRECT opening year → 1993

Reference CORRECT opening year → 1993, keeping the record ID

Rule	*Operation*	*Executed state*
AGREE	HOLD	✗ 1991 retained
ALLOW	CORRECT	✓ 1993
LITERAL	CORRECT	✓ 1993
WITNESS	CORRECT	✓ 1993

B Disputed residence
contrast set

Stored record: residence = Ulsan

Reading 1: "I live in Changwon now."
Reading 2: "I do not live in Changwon now."

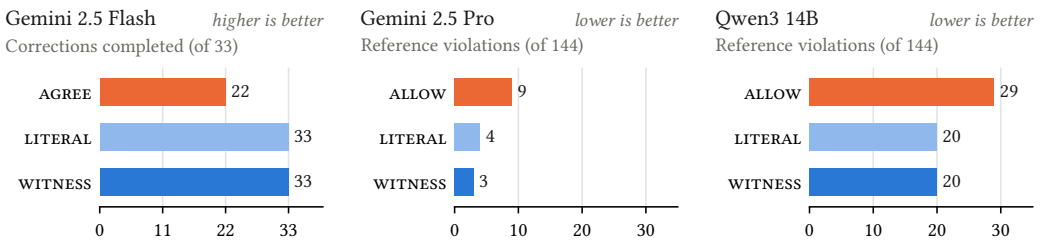
Shared proposal: CORRECT residence → Changwon; reply "You live in Changwon. Understood."

Reference HOLD: the meaning is unresolved, so Ulsan is kept

Rule	*Operation*	*Executed state*	*Reply matches state*
AGREE	HOLD	✓ Ulsan	✗
ALLOW	CORRECT	✗ Changwon	✓
LITERAL	CORRECT	✗ Changwon	✓
WITNESS	HOLD	✓ Ulsan	✗
REVIEW*	writes	✗ Changwon	—

*REVIEW generates its own proposal; its reply is not compared here.

Fig. H1. Detailed translated examples. A shows the supported opening-year correction. B shows why a residence value appearing in both readings can still lack affirmative support when one reading negates it. The latter belongs to the separate contrast source.



Same saved proposals within each panel. Baseline: AGREE for Flash, ALLOW for Pro and Qwen. Counts pool three repetitions of the 48 cases.

Fig. H2. Three component changes on fixed proposals. Gemini 2.5 Flash retains more correct corrections, Gemini 2.5 Pro admits fewer updates that violate the reference, and Qwen3 14B also reduces reference violations. Denominators and baselines differ as labeled; all panels are component comparisons.

I Admission Bounds, Operation Split, and Variant Audit on Saved Review Proposals

This appendix completes the REVIEW+gate replay of Section 6.3 with its two bounds, REVIEW+AGREE and REVIEW+ALLOW, and separates the gate's APPEND and CORRECT decisions. All five conditions use the same saved REVIEW proposals and the extractions from the same configuration, case and repetition (21 configurations × 48 cases × 3 repetitions). The two bounds were added after the LITERAL and WITNESS hybrids had been inspected. They apply the original rule definitions with the frozen executor and were scored once with the archived evaluator and reference: 6,048 new condition outputs, no model or provider call, no recomputation of the saved REVIEW, REVIEW+LITERAL or REVIEW+WITNESS scores, and no change in the hashes of the frozen source files. The comparison

is retrospective and reuses the authored cases; it adds no independent situations and no human judgment of the outputs.

Figure I1 summarizes the five conditions, and Table I1 restates how each condition treats the two operations. Tables I2–I4 give the endpoint totals, the decision counts by operation, and the attribution of paired changes; Table I5 splits the totals by the reference-defined case strata, Table I6 gives pooled paired intervals, and Table I7 lists the only five configurations on which the four rules differ. Table I8 and Figure I2 keep the per-configuration Review→Review+Witness results. Table I9 audits which of the 82 avoided reference violations meet the mechanical candidate-variant criteria.

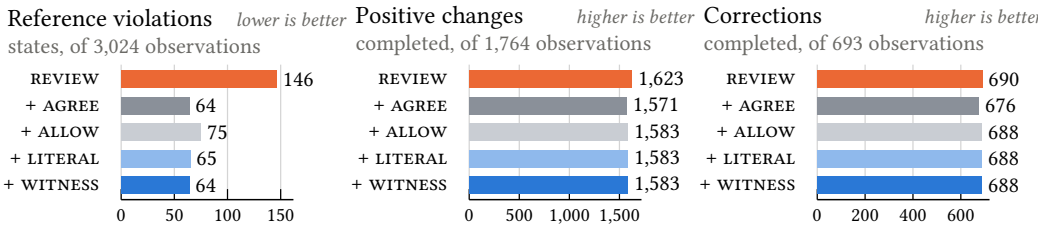


Fig. I1. Admission applied to saved Review proposals: the Agree and Allow bounds and the two evidence checks. The reference-violation endpoint follows the fixed state reference, including exact tuple mismatches. The same 21 configurations and all three repetitions appear in every bar; denominators are 3,024 for reference violations, 1,764 for positive changes, and 693 for corrections. Table I8 and Figure I2 report every configuration.

Table I1. Conditions on the same saved Review proposal. Extraction coverage, retention permission and the executor's checks are shared; an invalid extraction leaves the initial state and counts as an invalid output.

Condition	APPEND	CORRECT
Review	No gate; executor checks only.	No gate; executor checks only.
Review + Agree	Held unless the tuple is supported by every reading's fact list or already stored.	Same test as APPEND; held corrections form H when every reading is validly covered.
Review + Allow	As Agree.	Restores every correction in H.
Review + Literal	As Agree.	Restores a correction in H when $L_j = 1$ (Section 4.1).
Review + Witness	As Agree.	Restores a correction in H when $W_j = 1$ (Section 4.1).

Between the rules, every paired difference involves CORRECT only. Review+Allow, +Literal and +Witness each complete 12 more corrections and positive changes than Review+Agree; Review+Allow has 11 more reference-violating states than Review+Agree and Review+Witness, and 10 more than Review+Literal. Of the 23 corrections that Agree holds, Allow releases 23 (11 executed with a reference violation), Literal 13 (one) and Witness 12 (none); the 12 releases shared by all three rules are the ones that complete a required correction. In every observation, the four rules make the same APPEND decisions.

No condition produces a reference-violating state on the correction-eligible cases. On the new-information cases, the four rules coincide, and every reduction relative to Review (125 to 61) comes from APPEND holds. On the cases that require no change, APPEND holds reduce reference-violating

Table I2. Endpoint totals on saved Review proposals. The first five columns use all 3,024 observations (positive changes 1,764; corrections 693). The last three columns restrict all conditions to the 2,995 observations whose Review output and extraction are both valid, so that invalid extractions do not enter the comparison.

	All observations					Valid in every condition		
Condition	**Viol. ↓**	**Positive**	**Correct.**	**Invalid**	**Conform.**	**Viol. ↓**	**Positive**	**Correct.**
Review	146	1,623	690	21	2,849	146	1,619	688
Review + Agree	64	1,571	676	29	2,810	64	1,571	676
Review + Allow	75	1,583	688	29	2,811	75	1,583	688
Review + Literal	65	1,583	688	29	2,821	65	1,583	688
Review + Witness	64	1,583	688	29	2,822	64	1,583	688

Table I3. Original Review decisions by operation. The unit is an operation decision, and one observation can contain several; these counts are therefore not added to the observation-level totals of Table I2. "Unavailable" marks decisions in observations whose extraction was invalid. The last column counts executed operations that the reference marks as violating. Review has no gate, and in every condition 19 admitted appends were not executed by the shared executor.

Condition	**Operation**	**Proposed**	**Admitted**	**Held**	**Unavail.**	**Executed**	**Exec. viol.**
Review	APPEND	1,087	–	–	–	1,068	132
	CORRECT	762	–	–	–	762	14
Review + Agree	APPEND	1,087	977	108	2	958	61
	CORRECT	762	737	23	2	737	3
Review + Allow	APPEND	1,087	977	108	2	958	61
	CORRECT	762	760	0	2	760	14
Review + Literal	APPEND	1,087	977	108	2	958	61
	CORRECT	762	750	10	2	750	4
Review + Witness	APPEND	1,087	977	108	2	958	61
	CORRECT	762	749	11	2	749	3

Table I4. Paired endpoint changes relative to Review, attributed to the operation whose gate decision differed in the same configuration, case and repetition; "Inv." marks an invalid extraction. Relative to Review, no condition introduced a new reference violation or completed an additional positive change or correction, and no observation changed both operations. The attribution is descriptive and not an additive causal decomposition.

	Viol. prevented			Positive changes lost				Corrections lost		
Condition	**APP.**	**COR.**	**All**	**APP.**	**COR.**	**Inv.**	**All**	**COR.**	**Inv.**	**All**
Review + Agree	71	11	82	36	12	4	52	12	2	14
Review + Allow	71	0	71	36	0	4	40	0	2	2
Review + Literal	71	10	81	36	0	4	40	0	2	2
Review + Witness	71	11	82	36	0	4	40	0	2	2

Table I5. Outcomes by case stratum (cases; observations). The strata come from the frozen reference and were not inferred from model outputs: the 11 correction-eligible cases of the strict-correction endpoint, the other 17 positive cases, which require new information, and the 20 cases that require no positive change.

Stratum	Endpoint	Review	+ Agree	+ Allow	+ Literal	+ Witness
Correction-eligible (11; 693)	Viol.	0	0	0	0	0
	Correction	690	676	688	688	688
	Invalid	3	5	5	5	5
New-information (17; 1,071)	Viol.	125	61	61	61	61
	Positive	933	895	895	895	895
	Invalid	2	4	4	4	4
No change required (20; 1,260)	Viol.	21	3	14	4	3
	Conformity	1,223	1,237	1,226	1,236	1,237
	Invalid	16	20	20	20	20

states from 21 to 14 under every rule; the rules then diverge only through CORRECT, leaving 14 under Allow, 4 under Literal, and 3 under Agree and Witness.

Table I6. Pooled paired differences in percentage points with 95% case-cluster percentile intervals (5,000 draws over 48, 28 and 11 case clusters, keeping each case's three repetitions and 21 configurations together). The first two rows compare a hybrid with Review; the others compare rules on the same Review proposals. A zero marks identical paired outcomes. The intervals are exploratory and pointwise. They use a new fixed seed, so a paired difference identical to one in Section 6.3 can have interval endpoints that differ slightly from the archived ones.

Contrast	Viol. ↓	Positive ↑	Correction ↑	Conformity ↑
Review+Agree − Review	−2.71 [−4.30, −1.36]	−2.95 [−4.76, −1.47]	−2.02 [−3.32, −0.87]	−1.29 [−2.61, −0.10]
Review+Allow − Review	−2.35 [−3.90, −1.06]	−2.27 [−4.08, −0.79]	−0.29 [−0.72, 0.00]	−1.26 [−2.48, −0.30]
Allow − Agree	+0.36 [0.00, 0.93]	+0.68 [0.17, 1.36]	+1.73 [0.58, 3.17]	+0.03 [−0.66, 0.63]
Witness − Agree	0	+0.68 [0.17, 1.36]	+1.73 [0.58, 3.17]	+0.40 [0.07, 0.79]
Witness − Allow	−0.36 [−0.93, 0.00]	0	0	+0.36 [0.00, 0.93]
Witness − Literal	−0.03 [−0.10, 0.00]	0	0	+0.03 [0.00, 0.10]

Table I7. The five configurations whose Review proposals reach H, in the order Review / + Agree / + Allow / + Literal / + Witness. On the other 16 configurations, Agree holds no proposed correction, so the four rules coincide on every endpoint. Positive-change counts differ between rules only through the 12 corrections in this table.

Configuration	Held corrections (cases)	Viol. /144	Correction /33
Claude Sonnet 4	1 (CT22)	6 / 4 / 5 / 4 / 4	31 / 31 / 31 / 31 / 31
Claude Sonnet 4.5	4 (CT21, CT22)	9 / 3 / 7 / 3 / 3	33 / 33 / 33 / 33 / 33
DeepSeek 3.2	5 (CT21, CT22, CT46)	5 / 1 / 5 / 1 / 1	33 / 32 / 33 / 33 / 33
Gemini 2.5 Flash	12 (CT01, CT21, CT42, CT45, CT46, CT48)	5 / 1 / 2 / 1 / 1	33 / 22 / 33 / 33 / 33
Qwen3 14B (4-bit)	1 (CT11)	14 / 8 / 9 / 9 / 8	33 / 33 / 33 / 33 / 33

The 12 Gemini 2.5 Flash holds include 11 in the five correction cases of Section 6.1 (CT01, CT42, CT45, CT46 and CT48, with the same repetition counts) and one in CT21. CT21 and CT22 are

conflicting-transcript cases that require no change; their ten held corrections each produce a reference-violating state when released, and Literal and Witness both keep them held. CT11 is the Qwen3 14B destination overwrite discussed in Section 6.4, which Literal releases and Witness holds.

Every configuration. In Table I8, each arrow is Review → Review + Witness. Positive changes use 84 observations per configuration, corrections 33 and reference violations 144. Literal differs only by one additional reference-violating state for Qwen3 14B (nine rather than eight). On the four configurations where the gate prevents no reference-violating state (Claude Sonnet 4.6, Gemini 3 Flash Preview, 3.5 Flash-Lite, and 3.8 Flash), the hybrid loses 16 of the 40 positive changes, 14 through gate holds and 2 through invalid extractions; on Gemini 2.5 Flash-Lite and GPT 5.6 Terra, it prevents 11 and 9 reference-violating states at a cost of 7 and 1 positive changes, the latter due to an invalid extraction. Normalization leaves every configuration's corresponding Agree and Literal endpoint counts unchanged.

Table I8. All 21 configurations in the generator-preserving hybrid comparison.

Configuration	Viol. /144 ↓	Positive /84 ↑	Correction /33 ↑
Claude Opus 4.5	5 → 1	79 → 76	33 → 33
Claude Opus 4.6	8 → 2	76 → 75	33 → 33
Claude Opus 4.7	6 → 2	78 → 77	33 → 33
Claude Opus 4.8	3 → 1	81 → 80	33 → 33
Claude Opus 5.5	15 → 11	69 → 69	33 → 33
Claude Sonnet 4	6 → 4	77 → 75	31 → 31
Claude Sonnet 4.5	9 → 3	79 → 79	33 → 33
Claude Sonnet 4.6	3 → 3	81 → 81	33 → 33
DeepSeek 3.2	5 → 1	81 → 77	33 → 33
Gemini 2.5 Flash-Lite	14 → 3	69 → 62	33 → 32
Gemini 2.5 Pro	8 → 3	71 → 71	32 → 32
Gemini 3.1 Pro Preview	3 → 1	81 → 80	33 → 33
Gemini 3.5 Flash-Lite	0 → 0	84 → 75	33 → 33
Gemini 3.8 Flash	0 → 0	84 → 83	33 → 33
Gemini 3 Flash Preview	6 → 6	78 → 72	33 → 33
Gemini 2.5 Flash	5 → 1	82 → 81	33 → 33
GLM 5	3 → 0	81 → 80	33 → 33
GPT 5.6 Luna	7 → 4	77 → 77	33 → 33
GPT 5.6 Sol	13 → 6	71 → 71	33 → 33
GPT 5.6 Terra	13 → 4	71 → 70	33 → 32
Qwen3 14B (4-bit)	14 → 8	73 → 72	33 → 33

I.1 Variant Audit of the 82 Avoided Reference Violations

Section 6.3 asks which of the 82 reference-violating states that Review+Witness avoids relative to Review involve explicitly forbidden tuples and which meet mechanical criteria for candidate representational variants. The audit uses only archived artifacts. For each of the 82 paired observations, we took the reference components that the Review state failed from the saved scores and the held operation from the saved gate log of the hybrid. For each case, we collected the facts that are new relative to the initial state in every replayed output whose executed state conforms to the reference; in each affected positive case, all conforming states contain the same single new fact, the required fact. A held append counts as a *candidate variant* of that fact when it has the same relation

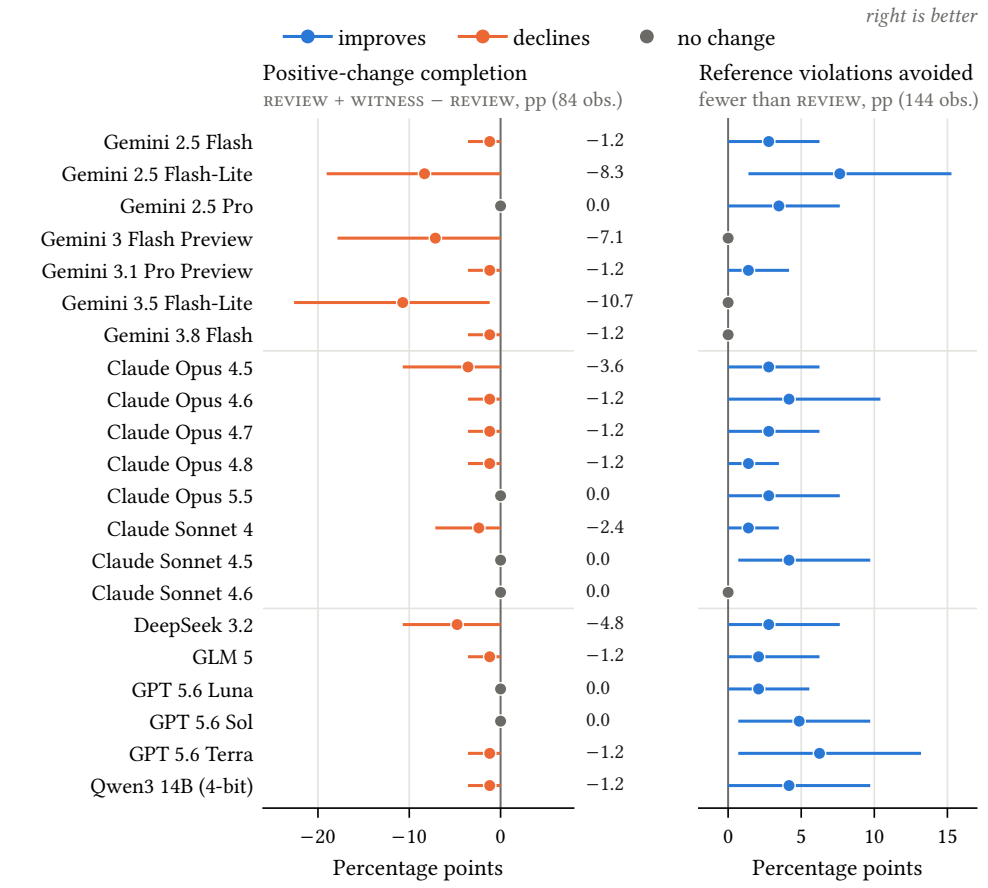


Fig. I2. REVIEW + WITNESS minus REVIEW, oriented so that improvement is to the right. Blue marks improvement, orange decline and gray no change; values at the right are point estimates in percentage points. Intervals cluster the three repetitions within each original case and are exploratory pointwise intervals. Both favorable restraint and unfavorable completion effects are shown.

and time and one of the following holds after Unicode NFC, case folding, and whitespace removal: subject and value are equal although the raw strings differ (spacing); the values are equal and one subject string contains the other or both denote the participant ("participant" or "p1"; subject label); or one value contains the other and the subjects match in the same sense (value form).

The rule was written after the avoided observations had been listed and is applied mechanically; it is not a semantic judgment, and no person rated the outputs. It does not identify potential variants requiring translation or restructuring, such as "email" for "이메일" or a subject that names the farm rather than the participant. Table I9 gives the result. Each of the 71 append observations contains exactly one held append, and in all 52 candidate-variant observations the exact required tuple is absent under both REVIEW and the hybrid, so positive-change completion fails either way.

The audit changes the reading of Table I4, not its counts. Of the 71 violations avoided by APPEND holds, 52 come from withholding a required fact written in another form, and 19 from withholding a fact that no conforming state contains; the 11 avoided by CORRECT holds all involve a forbidden value.

Table I9. Composition of the 82 reference-violating states avoided by Review+Witness relative to Review (observations; 21 configurations, three repetitions). "Forbidden" means that the Review state contains a tuple the reference forbids; all other append holds failed only because of an unlisted fact. Variant categories are mechanical candidate matches, not validated semantic equivalences. English glosses are translations.

Class (cases)	Required fact	Held tuple	Obs.
Append holds, variants of a required fact (52 observations, nine cases)			
Subject label (CT13)	준호 · hobby · 낚시	남편 준호 ("husband Junho"), participant의 남편 준호	10
Subject label (CT14)	순자 · birthplace · 임실	어머니 순자 ("mother Sunja"), participant의 어머니 순자	3
Subject label (CT15)	도윤 · workplace · 대한우편	아들 도윤 ("son Doyun"), participant의 아들 도윤	7
Subject label (CT16)	영숙 · learned_from · 이 사범님	동생 영숙 ("sibling Youngsook"), participant의 동생 영숙	8
Subject label (CT27)	participant · shop_name · 장미미용실	subject p1	1
Spacing (CT20)	participant · hobby · 우표수집	우표 수집 ("stamp collecting")	6
Value form (CT31)	participant · event_year · 2003년	2003; 서안농장 개업 2003년 and similar	11
Value form (CT44)	participant · hobby · 자전거	자전거 타기 ("riding a bicycle")	5
Value form (CT26)	participant · learned_from · 오 선생님	기타: 오 선생님	1
Other append holds (19 observations)			
Unlisted fact (CT16, CT24, CT26, CT29, CT31, CT39)	–	e.g., another teacher's name; email for 이메일; the farm as subject; subject and value swapped	17
Forbidden (CT23)	–	a destination on which the readings disagree (Qwen3 14B)	2
Correction holds (11 observations)			
Forbidden (CT11, CT21, CT22)	–	overwrite of a protected record with a forbidden value	11

J Follow-Up Runs

The runs in this appendix were collected after the results of the main article had been inspected, as tests of its diagnoses (main article, Sections 5.1, 6.1, 6.6, and 6.7). These were diagnostic follow-ups, not publicly preregistered independent replications. Each had a frozen contract that fixed the selection, prompts, structural schemas, comparisons, stopping rules, and analysis before any output. Each ran once, with no retry or repair; an invalid output counts as a system failure, and an unreturned call leaves its case out of that model's paired denominator rather than scoring it as zero. No model was trained, and no archived score was recomputed or overwritten. On the main cases, the reference is the authored specification of the main article; on DSTC2, it derives from the corpus labels. All paired intervals are 95% percentile intervals from 5,000 cluster bootstrap draws with seed 20261004, clustering by case on the main cases and by caller on DSTC2.

J.1 Revised Extraction Instruction

The comparison holds the saved controlled-path proposal, the Agree gate, the executor, and the reference fixed and varies only the extraction instruction. The revised instruction, written once before any response and not tuned afterward, adds the following paragraph to the original extraction

prompt (translated from Korean): "Record extraction locations explicitly. An atomic fact that a reading explicitly confirms after a correction must also be recorded in that reading's facts; do not omit it from the facts because the correction evidence is recorded in the correction witnesses. Record the corrected new value with the existing subject, relation, and time in the fact schema. Do not record negated, uncertain, or other people's facts as confirmed facts of the existing person. Do not copy a fact supported by only one reading into another reading's facts. Do not add facts absent from the source."

Gemini 2.5 Flash used temperature 0.2 and a thinking budget of 0, and Gemini 3.8 Flash temperature 0.2 and the LOW thinking level; both had an output cap of 8,192 tokens. For each of the five Gemini 2.5 Flash cases with extraction-induced holds, the selected repetition is the earliest one in which AGREE held the correction; each case received one new response under each instruction (Table J1). Under the original instruction, the four reproduced holds again show the pattern of the main article's Section 6.1: both fact lists are empty, and every reading carries an affirmed witness for the corrected value. Under the revised instruction, the witnesses are unchanged and the corrected tuple appears in both fact lists.

Table J1. Re-extraction of the five Gemini 2.5 Flash cases, one new response per instruction, gated by the unchanged AGREE rule on the saved proposal. "Shared" means that the corrected tuple is in the fact lists of both readings. No output in either arm produced a reference-violating state.

Case	Proposed correction	Rep.	Original: shared / completed	Revised: shared / completed
CT01	birthplace ← 대구 (Daegu)	1	yes / yes	yes / yes
CT42	birthplace ← 문경 (Mungyeong)	3	no / no	yes / yes
CT45	event_year ← 1993년	1	no / no	yes / yes
CT46	learned_from ← 신 선생님 (teacher Shin)	1	no / no	yes / yes
CT48	meeting_place ← 시민운동장 (civic stadium)	1	no / no	yes / yes
Total			1 / 1 of 5	5 / 5 of 5

CT01 recovered spontaneously under the original instruction and is therefore not counted as an effect of the revision. On Gemini 3.8 Flash, both instructions were applied to the saved controlled-path proposal of repetition 1 for all 48 cases. The outcomes are identical case by case: 11 of 11 strict corrections, 27 of 28 positive changes, 47 of 48 conforming states, and no reference violation under either instruction. A planned comparison on all 48 Gemini 2.5 Flash cases stopped at a provider error after 23 original-instruction and 27 revised-instruction extractions and is not analyzed; the five pairs in Table J1 were completed in two later collections, one of which stopped at a rate-limit error and was completed by a three-request recovery. Four of the ten responses in the table come from the stopped collection.

J.2 Learned Verifier

The verifier (Gemini 3.1 Pro Preview, temperature 0.2, LOW thinking level, output cap 8,192 tokens) received the public case (context, all readings, initial memory, and retention permission) and the indexed list of proposed APPEND and CORRECT operations of one saved REVIEW proposal, and returned, in a fixed JSON schema, one decision per operation: an approval flag and a short reason. A disapproved operation becomes HOLD; nothing else in the proposal changes, and the result is executed with the frozen executor and scored with the archived evaluator. The verifier saw neither

the extraction, the reference, nor any earlier gate decision. Its instruction reads (translated from Korean): "You verify memory writes in a personal conversation. For each APPEND or CORRECT proposal, judge whether the write is justified, considering all candidate transcripts, the dialogue context, the existing memory, and storage consent together. The transcript alternatives are possible readings of the same utterance, and the write needs support in all of them. Accept equivalent meanings that differ only in wording. Distinguish other people's information, differences in time, negation, uncertainty, and quotation from actual assertion. For CORRECT, check that it truly corrects the designated existing record; if there is an explicit correction, do not reject it only because it differs from the past value. Do not approve when storage consent is denied or unknown. No reference, earlier policy decision, or answer is provided, and the proposals may be wrong."

The selected proposals are repetition 1 of every case for Gemini 3.8 Flash and Qwen3 14B and for Gemini 2.5 Flash, except CT42, for which repetition 3 holds the correction (Table J1); the deterministic conditions are the archived scores for the same repetitions. All 144 calls returned valid decisions. The verifier held 4 of 15 proposed corrections and none of 17 appends on Gemini 2.5 Flash, nothing among 11 corrections and 17 appends on Gemini 3.8 Flash, and 7 of 18 corrections and 6 of 20 appends on Qwen3 14B (Table J2). The calls had a mean latency of 4.39 seconds (median 3.58) and used on average 673 input, 35 output, and 144 reasoning tokens.

Table J2. All conditions on the same saved Review proposals, one repetition per case. Denominators are 11 correction, 28 positive, and 48 cases for reference-violating and conforming states.

Configuration	Condition	Correct. /11	Positive /28	Viol. /48	Conform. /48
Gemini 2.5 Flash	Review	11	27	2	46
	Review + Agree	6	22	1	42
	Review + Allow	11	27	2	46
	Review + Literal	11	27	1	47
	Review + Witness	11	27	1	47
	Review + verifier	11	27	1	47
Gemini 3.8 Flash	Review	11	28	0	48
	Review + any of the four rules	11	27	0	47
	Review + verifier	11	28	0	48
Qwen3 14B (4-bit)	Review	11	24	4	42
	Review + any of the four rules	11	24	3	43
	Review + verifier	11	24	1	44

Relative to Review+Witness, the verifier differs only on Gemini 3.8 Flash (positive changes +3.57 percentage points, interval 0.00 to 10.71; conformity +2.08, 0.00 to 6.25) and Qwen3 14B (reference violations −4.17, −10.42 to 0.00; conformity +2.08, 0.00 to 6.25). Relative to Review+Agree on Gemini 2.5 Flash, it completes five more corrections (+45.5 points, 18.2 to 72.7), the same five that Allow, Literal, and Witness release. On Gemini 2.5 Flash, one of its held corrections, in the conflicting-transcript case CT21, reproduces the Review+Witness hold; the other three leave every endpoint as under Review. The three state differences from Review+Witness are described in the main article's Section 6.6.

J.3 Real Speech-Recognition Hypotheses (DSTC2)

The comparison with the main Korean source is not crossed: language, domain, reference construction, number and origin of readings, and sampling differ. A difference in holds cannot be attributed specifically to recognition noise.

Source and selection. The source is the official DSTC2 test release (https://github.com/matthen/dstc). One user turn was taken from each of 48 distinct dialogues, from 41 callers, by a hash-ranked rule with a fixed seed applied before any model output: 24 goal revisions, 12 denials without an explicit replacement, and 12 reference-enriched n-best turns whose recognition errors were identified with the labels, which form a separately reported diagnostic stratum. The sample serves diagnosis and does not estimate how often such turns occur.

Input and reference. The readings are all distinct non-empty strings of the live 10-best list in their original rank order (433 over the 48 turns); exact duplicates and blank strings are omitted without rewriting, and no likelihood weight is used. The history consists of the earlier system turns and top-ranked user hypotheses, and the previous official goal for food, area, and price range seeds the memory; the current transcript and labels are withheld. The domain adapter replaces the Korean personal relations with these three slots and leaves the gate logic unchanged; retention permission is set to allowed as a task fixture, not as observed consent. The reference is the official goal derived from human-reviewed semantic labels, compared as exact lowercase values without normalization. Goal conformity requires all three slots to match; a new wrong write introduces a slot value absent from the official goal, whereas a stale prior value fails goal conformity without counting as a new wrong write; required-change completion counts turns whose goal changes. Actions and consent are not scored. DELETE is not supported by the adapter, and no selected turn required removing a slot.

Collection. Gemini 3.1 Pro Preview and Gemini 3.8 Flash ran with temperature 0.2, the LOW thinking level, an output cap of 8,192 tokens, and one repetition. A model stopped at its first transport or provider failure. Gemini 3.1 Pro Preview completed all 48 turns; Gemini 3.8 Flash stopped after 22 valid turns (21 callers), with no denial turn among them. The 142 attempts produced 70 valid pairs. The two models are not pooled.

Table J3. DSTC2 outcomes by stratum, as goal conformity / required changes completed / new wrong writes. "Ungated" executes the proposal without admission. Gemini 3.8 Flash is a partial run and is not pooled with Gemini 3.1 Pro Preview.

Model and stratum (turns)	**Ungated**	**Allow**	**Agree**	**Literal**	**Witness**
Gemini 3.1 Pro Preview					
Goal revision (24)	20 / 20 / 1	20 / 20 / 1	5 / 5 / 0	5 / 5 / 0	5 / 5 / 0
Denial (12; 1 change)	10 / 1 / 2	10 / 1 / 2	12 / 1 / 0	11 / 1 / 1	12 / 1 / 0
Recognition error (12; 11 changes)	3 / 2 / 0	3 / 2 / 0	1 / 0 / 0	1 / 0 / 0	1 / 0 / 0
All (48; 36 changes)	33 / 23 / 3	33 / 23 / 3	18 / 6 / 0	17 / 6 / 1	18 / 6 / 0
Gemini 3.8 Flash, partial					
All valid (22; 21 changes)	9 / 8 / 1	9 / 8 / 1	3 / 2 / 0	3 / 2 / 0	3 / 2 / 0

Holds and paired comparisons. On Gemini 3.1 Pro Preview, Agree held a proposed correction in 20 outputs (16 goal revisions, two denials, and two recognition-error turns) and held no append. Ten holds arise from different values extracted from different hypotheses and ten from a value supported by only some hypotheses or omitted from some; none shows the extraction pattern of the main source. Allow releases all 20: 17 complete a required change and three introduce a wrong value (DSTC2_17, DSTC2_40, and DSTC2_45). Literal releases only DSTC2_45, a wrong write, and Witness releases none. Relative to Allow, Witness changes goal conformity by −31.25 percentage points (caller-cluster interval −45.83 to −16.33), required-change completion by −47.22

(−62.86 to −32.26; 36 turns from 31 callers), and turns with a new wrong write by −6.25 (−13.46 to 0.00); LITERAL changes them by −33.33 (−47.73 to −19.23), −47.22, and −4.17 (−10.42 to 0.00). On the partial Gemini 3.8 Flash run, AGREE held seven corrections, six for partial support or omission and one for different values.

Data terms. DSTC2 is used under its own terms. The archive contains no corpus text, transcript-bearing input, gold value, or model response derived from it; it provides the selected dialogue paths, turn indices, and source hashes, the adapter and evaluation code that reconstruct the 48 inputs from an independently obtained copy, and scalar outcome flags with pseudonymized caller clusters that reproduce the tables and intervals.

J.4 Korean Memory-Operation Data

The Korean memory-operation data of Bae et al. (main article, reference on memory management in long-term conversations) would supply an externally labeled reference for the specification confound. The official public repository we inspected contains Korean samples and machine-translated English material, but not the original operation-label pairs, example identifiers, session links, or official split files that a replication requires. No model call or evaluation was therefore made on this source. Its REPLACE operations also include changes over time and added detail, so not every REPLACE is a correction of an earlier error.

K Variant-Tolerant Re-Scoring

K.1 Definition and Implementation

The variant-tolerant reference re-scores the saved final states; it executes no policy and calls no model. A final tuple f matches a reference tuple r when $f = r$, or when relation and time are equal, the subjects match, and the values match. After Unicode NFC, case folding, and removal of all whitespace, subjects match when they are equal, when both are participant labels (`participant` or `p1`), or when one contains the other and neither is a participant label; values match when they are equal or one contains the other. A required tuple is present when some final tuple matches it, a final tuple that matches a required tuple is not an extra fact, and a protected record is preserved when its final tuple matches its initial tuple. Forbidden tuples, obsolete versions, removals, actions, and validity are scored exactly as before, so the second reference can only credit an output. The scorer wraps the archived evaluator: with matching switched off, it reproduces every score field except the call-identifier string in all 42,336 archived condition rows of the panel (3,024 episodes, 14 conditions), and the reference-violation and correction fields of the 1,152 archived controlled-path ALLOW rows. The rule was written after the variant audit of Appendix I and is broader than that audit only in applying to every final tuple rather than to held appends. The scripts are in the archive under `analysis`.

Table K1 lists every pair that the rule accepts. The accepted candidate pairs differ from the required fact by a kinship label, an underscore or possessive form of the same subject, the speaker identifier `p1` for the participant, a spacing difference, the year without its counter suffix (2003 for 2003년), or additional words around the same value. No forbidden tuple is affected, because forbidden tuples are not relaxed.

Acceptance by this matcher is not an independent semantic judgment. Substring containment does not establish that subject labels identify the same entity or that additional words preserve the fact's meaning. The 38 pairs are candidates for contextual human validation, not a gold equivalence set. The alternative reference is an exploratory sensitivity analysis and does not bound the possible outcomes of a future human evaluation.

Table K1. All 38 candidate surface-variant pairs mechanically accepted by the variant-tolerant reference, grouped by case. The third column shows the differing subject or value of each accepted tuple and, in parentheses, the number of condition-level observations among the 42,336 scored rows in which it occurs. English glosses: 남편 husband, 어머니 mother, 아들 son, 동생 younger sibling, 기타 guitar, 개업 opening, 타기 riding.

Case	Required fact	Accepted forms (observations)
CT13	준호 · hobby · 낚시	남편 준호 (36); participant의 남편 준호 (27); 남편_준호 (1); participant_spouse(준호) (1); 준호 (participant의 남편) (1); participant_spouse_준호 (1)
CT14	순자 · birthplace · 임실	어머니 순자 (215); participant의 어머니 순자 (29); participant's mother 순자 (7); 순자(어머니) (2); participant_mother_순자 (1); 순자 (participant의 어머니) (1)
CT15	도윤 · workplace · 대한우편	아들 도윤 (75); participant의 아들 도윤 (16); participant_son_도윤 (2); 아들_도윤 (1)
CT16	영숙 · learned_from · 이 사범님	동생 영숙 (95); participant의 동생 영숙 (15); participant_sibling:영숙 (1); participant_sibling_영숙 (1); 영숙(p1 동생) (1); participant.younger_sibling.영숙 (1)
CT20	participant · hobby · 우표수집	우표 수집 (29)
CT26	participant · learned_from · 오 선생님	오 선생님 (기타) (4); 기타: 오 선생님 (4); 기타:오 선생님 (2); 오 선생님(기타) (1); 기타 - 오 선생님 (1)
CT27	participant · shop_name · 장미미용실	p1 (115)
CT30	participant · travel_destination · 여수	p1 (12)
CT31	participant · event_year · 2003년	2003 (348); 서안농장 개업 2003년 (11); 농장 개업 2003년 (3); 서안농장 개업: 2003년 (3); 2003년 서안농장 개업 (2); 농장을 처음 연 해는 2003년 (2); 2003년 농장 개업 (1)
CT44	participant · hobby · 자전거	자전거 타기 (10)

K.2 Rule Comparisons and Configurations

Table K2 repeats main-article Table 6 with Qwen3 14B and adds the violation counts under the variant-tolerant reference. For the completed replays reported in that table, strict correction counts are identical under both references, and every observed difference between two rules is the same; only the level of the violation counts falls. Across the full panel, controlled-path strict corrections are 673, 685, and 685 under Agree, Literal, and Witness under both references, and violations fall from 131, 133, and 131 to 54, 56, and 54. On the 36 controlled-path outputs and the 23 review-proposal outputs on which at least two of the four rules produce different endpoints, every available pairwise output-level difference is the same under both references. These two counts come from scores, not from the gate logs. The 23 review-proposal outputs are the exposed set. The 36 controlled-path outputs lie within the 49 exposed outputs of Table G5: Allow was not replayed on Claude Sonnet 4 and 4.5, DeepSeek 3.2, and GLM 5, whose 12 kept holds the other three rules all keep, and on one further exposed output Allow's release changes no endpoint under either reference. Fourteen exposed controlled-path outputs lack an Allow replay. The available comparisons therefore do not establish invariance for every rule pair on every panel output. Human validation has to use the exposed outputs, after completing the missing replays (Section 9 of the main article). The script `analysis/exposure_sets.py` recomputes the exposed set from the gate logs, the outputs on which the replayed rules leave different executed states, and the score-based set, checks that each contains the next, and lists the review scope. By contrast, Review and Review+Witness produce different endpoints on 127 outputs, and their difference changes with the reference on 52; the controlled path and Review differ on 199 outputs, and on 61 the difference changes.

Table K2. Rules on identical controlled-path proposals, in the order Agree / Allow / Literal / Witness, three repetitions. Strict correction is over 33 observations and reference-violating states over 144.

Configuration	Correction (both)	Viol., exact	Viol., variant-tolerant
Gemini 2.5 Flash	22 / 33 / 33 / 33	3 / 3 / 3 / 3	2 / 2 / 2 / 2
Gemini 2.5 Pro	31 / 31 / 31 / 31	3 / 9 / 4 / 3	0 / 6 / 1 / 0
Gemini 2.5 Flash-Lite	31 / 31 / 31 / 31	16 / 19 / 16 / 16	10 / 13 / 10 / 10
Gemini 3 Flash Preview	33 / 33 / 33 / 33	12 / 12 / 12 / 12	9 / 9 / 9 / 9
Gemini 3.1 Pro Preview	33 / 33 / 33 / 33	2 / 2 / 2 / 2	1 / 1 / 1 / 1
Gemini 3.5 Flash-Lite	32 / 32 / 32 / 32	7 / 12 / 7 / 7	4 / 9 / 4 / 4
Gemini 3.8 Flash	33 / 33 / 33 / 33	1 / 1 / 1 / 1	0 / 0 / 0 / 0
Qwen3 14B (4-bit)	33 / 33 / 33 / 33	20 / 29 / 20 / 20	17 / 26 / 17 / 17

Table K3 gives every panel configuration in the order Review / Review+Witness / controlled path. Under the exact reference these are the counts of Tables G1 and I8; under the variant-tolerant reference, Review has no reference-violating state on 10 configurations, so the gate prevents nothing there while it still holds appends.

Table K3. All 21 configurations under both references, in the order Review / Review+Witness / controlled path. Reference violations are over 144 observations and positive changes over 84.

	Viol. /144 ↓		Positive /84 ↑	
Configuration	**Exact**	**Variant-tol.**	**Exact**	**Variant-tol.**
Claude Opus 4.5	5 / 1 / 6	0 / 0 / 3	79 / 76 / 78	84 / 77 / 81
Claude Opus 4.6	8 / 2 / 6	2 / 2 / 4	76 / 75 / 78	82 / 75 / 80
Claude Opus 4.7	6 / 2 / 3	0 / 0 / 0	78 / 77 / 80	84 / 79 / 83
Claude Opus 4.8	3 / 1 / 2	0 / 0 / 0	81 / 80 / 82	84 / 81 / 84
Claude Opus 5.5	15 / 11 / 12	0 / 0 / 0	69 / 69 / 72	84 / 80 / 84
Claude Sonnet 4	6 / 4 / 6	1 / 0 / 0	77 / 75 / 76	82 / 79 / 82
Claude Sonnet 4.5	9 / 3 / 3	4 / 0 / 0	79 / 79 / 81	84 / 82 / 84
Claude Sonnet 4.6	3 / 3 / 3	0 / 0 / 0	81 / 81 / 81	84 / 84 / 84
DeepSeek 3.2	5 / 1 / 7	4 / 0 / 3	81 / 77 / 76	82 / 78 / 80
GLM 5	3 / 0 / 2	0 / 0 / 0	81 / 80 / 81	84 / 80 / 83
GPT 5.6 Luna	7 / 4 / 4	0 / 0 / 0	77 / 77 / 80	84 / 81 / 84
GPT 5.6 Sol	13 / 6 / 8	0 / 0 / 0	71 / 71 / 75	84 / 77 / 83
GPT 5.6 Terra	13 / 4 / 5	1 / 0 / 1	71 / 70 / 76	83 / 74 / 80
Gemini 2.5 Flash	5 / 1 / 3	3 / 0 / 2	82 / 81 / 80	84 / 82 / 81
Gemini 2.5 Flash-Lite	14 / 3 / 16	8 / 0 / 10	69 / 62 / 56	75 / 65 / 62
Gemini 2.5 Pro	8 / 3 / 3	1 / 0 / 0	71 / 71 / 73	78 / 74 / 76
Gemini 3 Flash Preview	6 / 6 / 12	6 / 6 / 9	78 / 72 / 71	78 / 72 / 74
Gemini 3.1 Pro Preview	3 / 1 / 2	3 / 1 / 1	81 / 80 / 81	81 / 80 / 82
Gemini 3.5 Flash-Lite	0 / 0 / 7	0 / 0 / 4	84 / 75 / 74	84 / 75 / 77
Gemini 3.8 Flash	0 / 0 / 1	0 / 0 / 0	84 / 83 / 83	84 / 83 / 84
Qwen3 14B (4-bit)	14 / 8 / 20	11 / 5 / 17	73 / 72 / 66	76 / 75 / 69

K.3 Decomposition and Intervals

The decomposition of main-article Table 8 is an identity on paired observations: for each configuration, case, and repetition, the controlled path minus Review equals the controlled path minus

Review+Witness plus Review+Witness minus Review. The two left-hand conditions share the extraction and the Witness gate and differ only in the proposal; the gate term is attributed to the held operation recorded in the saved gate log, and to invalid extraction when the hybrid output is invalid. No paired observation changed through both a held APPEND and a held CORRECT. Table K4 gives pooled paired differences with 95% case-cluster percentile intervals (5,000 draws, seed 20261005, keeping all configurations and repetitions of a case together). Because the seed differs from the archived analyses, the exact-reference gate intervals differ slightly from those of Appendix I.

Table K4. Pooled paired differences in percentage points with 95% case-cluster intervals over 48 (violations, conformity) or 28 (positive changes) case clusters.

Contrast and reference	**Viol. ↓**	**Positive ↑**	**Conformity ↑**
Proposal: controlled path − Review+Witness, exact	+2.22 [+1.03, +3.67]	+0.96 [−0.79, +2.72]	+0.56 [−0.63, +1.79]
Proposal: controlled path − Review+Witness, variant-tolerant	+1.32 [+0.56, +2.22]	+2.49 [−0.11, +5.44]	+1.46 [−0.26, +3.37]
Gate: Review+Witness − Review, exact	−2.71 [−4.33, −1.36]	−2.27 [−4.08, −0.85]	−0.89 [−2.15, +0.26]
Gate: Review+Witness − Review, variant-tolerant	−0.99 [−1.69, −0.40]	−5.22 [−8.22, −2.49]	−2.61 [−4.79, −0.73]
Complete pipeline: controlled path − Review, exact	−0.50 [−1.92, +0.93]	−1.30 [−3.40, +0.79]	−0.33 [−1.88, +1.26]
Complete pipeline: controlled path − Review, variant-tolerant	+0.33 [−0.66, +1.32]	−2.72 [−4.59, −1.02]	−1.16 [−2.68, +0.40]